

A Briefed Review on Phishing Attacks and Detection Approaches

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Abstract—Phishing is contemplated to be one of the most deadliest threats to internet users and still continues to grow in an exponential rate. This has resulted the hackers to generate more creative attacks and execute them on a large scale. The working concept of a phishing attack is to create the exact replica of an original site and direct users to land on the sites page. Due to the sites page being fake and similar to the original one, legit users are generally fooled to perform activities on such pages. Phishing is an attack that is mostly executed by combining fundamentals of social engineering with evolving technical methods. This cloaking of the fake site to function as the real one convinces the user to reveal their personal details such as their bank accounts and respective passwords. Hence, in recent times a comprehensive study on past attacks is mandatory to prepare ourselves from being a victim to such threats. This research paper puts forward a brief discussion on previous and existing attacks to provide a better understanding on the working principles of such threats s to encourage the development of anti-phishing techniques in the future. The primary objective of the presented work is to generate awareness amongst individuals and educate them from such attacks. Further, this review aims to assist policy makers and software developers in making right decisions and creating a virus free environment.

Keywords— Detection techniques, Phishing attacks, Prevention approaches, Security threats

I. INTRODUCTION

Phishing attacks are generally accompanied with the most widely used social engineering attacks that are implemented using advanced technology for attacker's personal gain. The hacker uses this trick and puts the user in a vulnerable situation of revealing his identity. The users being unaware of this attack are exposed to being victims; thereby reveal their financial statements, mobile OTP's and mail ID passwords. Such kinds of attacks are made on targeted users with the intention of political gains [1]. This attack has been witnessed to tremendously increase with the sudden rise of internet services being acquired in every household. To overcome this rising issue, the department of information security has designed and developed standard procedures and protocols to safeguard online information sharing and money laundering. With this, every organization has also set its security measures to not only avoid being under the attack situation, but also preventing the servers from future attacks [2]. In such a situation, working employees of an organization yearly review their system flaws and generates prevention strategies. Still, there are limitless forms of attack that tricks individuals into revealing their personal information, which are achieved through social engineering. Phishing attack is the most widely implemented of all. The attack works on baiting the user with fake mails

that seem like the ones genuinely send by bank accounts. Apart from this, users also receive cash winning mails that can be activated through clicking certain links. And once the user clicks on the link being provided, he is directed to the home page of the fake website on social networking sites [3]. Such baiting techniques are commonly used by high-end attacker that forces the victims to provide their card numbers to capture user details. This process of spoofing mails is carried out all across the net with thousand targets, in a hope that at least few might fall in the trap. These spoofed mails are intentionally designed to look professional so that the users can easily be fooled. Therefore, it is said that the primary aim of this attack is to focus on individuals with little or no knowledge of online transfers; as such individuals might think that the response is coming from a legit organisation; thereby giving an advantage of the users weakness to the attacker [4].

For instance, phishing attacks were first experienced in the year 1995 wherein the users were compelled to share their AOL information publically. Since then, this attack has always been on the rise and major developments have been made to trap the users in a fool-proof way [5]. In recent years it has been brought to notice that the phishers have been focusing on webmail attacks in comparison to other social engineering attacks [6]. A total of 33 percent webmail attacks were accounted in the year 2018, followed by 29 percent of attacks being made on security organisations. With regards to certain financial aspects of phishing, it has been found that "gift cards" have also been used to encash money from users. This kind of attack has been proved to occur at a minimal stage. However, FBI has estimated around 26,371 amounts of people being suffered as victims under such scams [7].

An approximate value of 100 million dollars has been registered as the net loss of industries such as health cares, BFSI and educational institutions [8]. The scam included, sending web-mails to working employees to gain their login credentials. Once the phishers got access to this information, they further used this to gain access to their payroll systems, and executed codes so that the employees can no longer receive notifications about changes being made in their accounts. Later on, the phishers change the employee's card details and channel employees funds into their own account [9].

The scams of phishing were not only limited to health care industries but were also observed on online gaming platforms. These scams were executed to steal login credential of players by offering them a "free skin gateway" that could only be accessed by clicking on a link that directed them to the phishing site. The players were put in an

unknown vulnerable situation that made them to enter their details by scrolling on a fake chat bar [10]. The process was carried out as the replica of the original site so that the players get an impression of legitimacy. The gaming players were prompted to login via an app called as Steam, which was the fake generated site in reality. The entered credentials of the users were captured and a two way authentication technique was followed, to assure them with authenticity.

This attack also served the purpose of creating one of the most devastating situations in Ukraine in 2015; wherein the administrative department that handled power distribution of the country was targeted with spear phishing attacks. This attack involved a Word document that contained virus and was triggered on downloading. This little act by software developers led to the instalment of malicious files on the server system and the entire organisation experienced a shutdown for 23 hours, resulting in power failure throughout the country. This attack however displayed an exemplary example of how a constructively planned phishing attack can be.

The above conducted survey indicates the utmost need to comprehend the extent of such attacks and develop techniques to combat them. Therefore, this study presents a summarised review on existing attacks and mention ways on how to prevent them from occurring.

The rest of the paper is segmented as follows: section 2 discusses the existing forms of phishing attacks followed by literature survey in section 3. A detailed module on prevention strategies is summed in section 4 proceeded by detection techniques in section 5. The paper is finally brought to an end in section 6.

II. TYPES OF PHISHING ATTACKS

The working concept of a phishing attack is to trick the user in to performing certain actions that are against his own interests. A total of 12 phishing attacks are described in this module along with a summarized description on how to identify them.

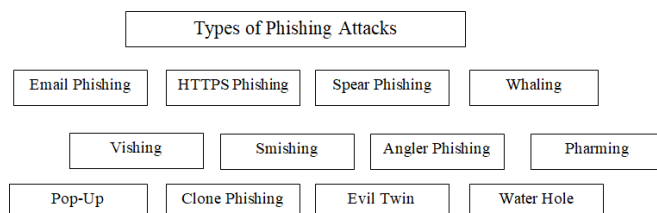


Fig. 1: Types of Phishing Attacks

A. Email Phishing

This phishing attack is the most widely used attack in the cyber-world. The intent of the malicious actor is to send mails to a user impersonating as a genuine brand that leads to individuals either clicking on the link or downloading a file. The intention of the hacker is to steal personal credentials of the user and install malicious code files on the server of the end user. However, the most legit way to identify this attack is to have a keen eye on shortened links and constantly verify for spelling errors. Attackers also make use of illegitimate domains for creating their mail ID's.

B. HTTPS Phishing

After the rise in increase of such attacks, organisations began to make use of HTTPS instead of just HTTP in order to increase the legitimacy of the site. But unfortunately, hackers have made their way out to overcome this security and inject codes of virus directly into the HTTPS web mails. Hence, it's necessary to verify the link before clicking on it. This can be done by identifying shortened links and links that are embedded in texts.

C. Spear Phishing

This attack also takes place through web-mails but is however different from email phishing. Phishers in this form of attack gathers information of individuals from publically available datasets such as social sites or company sites and further targets them with emails using real names from the organisation. This makes the targeted individual believe that he has received a mail from someone within the organisation; resulting in quick actions taken in response to the mail. However, this form of phishing can be identified by looking on unexpected mails received from different departments of an organisation.

D. Whaling

This form of attack takes place on a corporate level, wherein the malicious actor makes use of corporate website to gain information of the senior most members. The attacker further impersonates himself as a person belonging to the same hierarchy of leadership and asks for requests such as money transfer. The only way to identify such frauds is to make sure that no mails of such money transfer requests are being sent on personal ID's.

E. Vishing

This attack is derived from voice phishing wherein the attacker makes use of a phone call that creates a sense of urgency and leads an individual to take an action. However, sensing that the number belongs to some unusual location is the only way to identify this attack.

F. Smishing

This attack involves sending a text to an individual that forces him to take an urgent action that falls against his own interest. In such a scenario its always safe to review the area code before responding to such mobile messages.

G. Angler Phishing

This attack is generally observed to be the combination between smishing and vishing and takes place on social media platforms; wherein the hackers sends s direct message or a voice note to the targeted individual that forces him to take the necessary action. Hence, it's always advised to make sure that the social media user is a legit person and not an impersonated actor.

H. Pharming

This is the most difficult attack to detect as the malicious actors in this attack inject codes of virus directly onto the DNS server. This server is responsible to convert the URL so entered into the IP address of the legit site. Hence, when an attacker makes use of this attack, he adjusts the URL and redirects it to the IP address of the fake site so created,

making it difficult for the user to distinguish between the two. It's difficult for an individual to recognise such attacks but however, he needs to be careful with the websites colour and strange fonts.

I. Pop-Up Phishing

Pop-ups are considered to be those notification boxes that appear when a person visits a site. Malicious actors in such a case inject codes in the blocks of pop-up ads, forcing an individual to click on them. Once the individual clicks on the "Allow" button, codes are automatically installed on the users end. Therefore, shifting to full screen mode is the only way to avoid such pop-up ads.

J. Clone Phishing

Malicious users generally keep a track of the normal clicking activities of an individual and makes use of these services to leverage an individual to acquire them. Attackers research onto the areas in which services an organisation invests in and further targets them with those specific mails. The only way to identify this attack is to look out for emails that ask for personal information to provide regular services.

K. Evil Twin

This attack takes place on hotspots, wherein the attacker creates a fake Wi-Fi hotspot and provides access to individuals and further gains their credentials. The attacker in such a scenario engages in man-in-middle attack and performs the evil twin. Users must always look out for hotspots that do not ask for credentials before logging in.

L. Water Hole Phishing

This attack is also related with the corporate individuals, wherein the attacker keeps a track of the websites a company uses and further changes the IP address of that site to the fake site. Being unaware of this act, the company's employee visits the site and becomes a victim to this attack by downloading malicious files. Restricting browser rules is one of the ways to avoid this attack. The other way is to make use of security firewalls that prevents the system from such server breaches.

III. LITERATURE SURVEY

Phishing attacks are no less than a major crime in the cyber world, as the attacker executes such attacks by manipulating the personal information of a user data. Hence, this concern has become a major security issues in corporate world and bigger organisations that are responsible for security services. Where multiple techniques to attack a server system still exist, there many toolbars are also available on various internet browsers to warn users about phishing sites. Below are some of the research works performed by multiple authors that have contributed their study to detect such attacks and design methodologies to prevent them.

"Kiran and Sunil Pawar. (2020)" in [11] presented various types of phishing techniques along with respective detection algorithms. The authors observed that the cyber chain continued to expand because malware actors took the advantage of the user's weaknesses and focused to target only those individuals who had less knowledge with regards to the working of sites asking for login credentials. Their

research work also included of how phishers targeted an organisation's security by targeting the vulnerability of the servers system. Further, the researchers presented a comprehensive review on how such attacks can be detected and resolved using machine learning algorithms.

"Rana Alabdan.(2020)" in [12] gave a detailed study on attacks, methods and techniques to detect phishing threats in a server system. The review enhanced the development and focus of software developers towards the creation of anti-phishing techniques. The authors discussed in detail of how a planned attack can cause an entire shutdown of an organisation. Through their research they tried to educate developers and generated an awareness of how the virus may contaminate a system without the knowledge of the user. Overall, the paper encourages inclining towards anti-phishing techniques to avoid the situation of being under attack.

"Rastenis and Janulevicius. (2020)" in [13] proposed a taxonomy of a much broader classification techniques for phishing attacks. The authors focused only on web-mail attacks that were responsible to gain access to user's personal information without their consent. Their research work included the drawbacks of the existing system and how they could be enhanced to further avoid the situation of phishing attacks. The study also gives a detailed description on multiple categories of attacks and describes those attacks in six phases. The taxonomy of comparison that the authors performed was proven to be better than the existing classification techniques.

"Tavella and Merlo. (2020)" in [14] suggested a mechanism to develop those systems that were prone to weak security frameworks. The authors tried to focus on phishing attacks taking place on android phones wherein the attacker makes use of vishing attacks and tries to gain access to user credentials. The work also contributed in developing respective API's to avoid middle – man – attacks. The authors proposed a convenient method to show how certain features contribute towards mobile phishing attacks such as password managers on instant apps.

"Kathrine and M. Praise. (2019)" in [15] described variants of phishing attacks. Their work consisted of how threat actors could steal personal information of PAN card and adhar cards by tricking an individual to enter their personal details. Further, the authors implemented a model based on supervised and unsupervised learning techniques in order to detect the presence of unknown malwares in a software system. The authors also focused on how grieved the issue of malware attacks are and aimed to create awareness amongst individuals from being victims to such attacks.

"Anthony, and S. Bhelwa. (2019)" in [16] presented their views on attack challenges as online transaction take place on an everyday basis. They paper also conveys of how innocent people fall in this trap unknowingly. The authors focused on how machine learning algorithms assisted to detect the presence of malware attacks in the system. They also presented techniques and approaches to prevent such attacks in the future.

"Aleroud and L. Zhou. (2017)" in [17] proposed a taxonomical approach that focused on countermeasures and vector outputs of a phishing system. The paper further gave details on how the weaker systems in an organisation were

exposed to such attacks. The primary purpose of this research work was to provide a direction to software developers to design and creates anti phishing strategies that could prevent such attacks. The paper described on attacks taking place on social sites and provided machine learning methods to evaluate and detect the malware present in the server system.

The constructed table below gives a survey on machine learning approaches used by various scholars.

Table 1: Research study based on classification algorithms

Authors	Contribution	Weaknesses	Algorithms
K. A. Moul [18]	Highlighted steps to fight against phishing attacks	Small size dataset	SVM classifier
G. V. Bochmann [19]	Proposed a method to detect DOM attacks	Low level Recall measure	Bayesian Statistics
T. Wu [20]	Implementation of OCR technique to detect mobile attacks	Low level of accuracy as compared to other techniques	Semantic ontology
S. M. Banu [21]	Briefed different techniques and types of phishing attacks	Review on less survey was conducted	NIL
A. Pardeshi [22]	Presented phishing strategies with 100 percent accuracy	Small dataset was experimented upon	Random Forest and SVM classifiers
K. S. C. Yong [23]	Review on current and past approaches	Review on less survey was conducted	NIL

IV. ANALYSIS OF DETECTION TECHNIQUES

All the existing conventional approaches to detect phishing attacks are capable of detecting only 20% of the attacks being made on the systems server. Also these detection approaches performed with low accuracy. Therefore, there was an urgent need to develop strategies and techniques to overcome the existing limitations and challenges. However, in recent years the authors have come up machine learning techniques that produced optimized results with enhanced accuracy. Despite the fact that the process is time consuming and generally works on a smaller dataset, this method of detecting attacks using ML strategies is commonly adopted by software developers. Along with ML, certain techniques using the concepts of deep learning

are also exhibited. Techniques under deep learning generally involves the implementation of CNN architectures, as they provide easy execution of detection and classification strategies using multiple layers of the architecture involved. Apart from the execution of ML and DL techniques, heuristic based strategies are also employed to detect the same. All the research works in the existing studies focus on adapting to strategies that might help increase the accuracy of the overall system. But however, not many researchers have succeeded in accomplishing this task.

This section gives a briefed analysis of three detection techniques used to analyse a phishing attack.

A. Machine Learning based Detection Technique

One of the major advantages of using machine learning concepts is its ability to detect and classify labels accurately until an optimised model is achieved. In phishing detection, the role of ML strategies to target websites that is responsible to inject such attacks. In order to perform this procedure the dataset obtained must be trained in such a manner that it is capable of detecting all the features related to the phishing site. Further, ML adapts to multiple techniques of using specific classifiers and feature selection methods that would work to analyse the attacks. Below depicted is the conventional workflow of an ML for malware detection and classification.

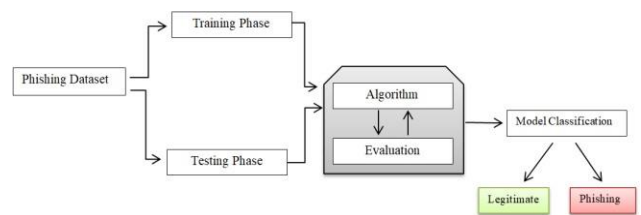


Fig. 2: Workflow of ML Classification

Authors in [24] carried out phishing detection techniques using Random Forest as a classifier along with a combination of binary classifiers. The authors acquired the dataset from Mendeley site and executed the detection algorithm on trained RF technique. The feature selection algorithms so used resulted in selecting efficient features and were further classified using the same algorithms. “A., Jhanjhi, N., & Supramaniam. (2019)” in [25] implemented the detection technique by combining bagging, boosting and stacking methods. This model resulted in the formation of ensemble learning model that was capable to extract 30 features from the dataset. The dataset however was obtained from the UCI repository and further implementation took place to accomplish the desired accuracy.

Authors in [26] implemented detection techniques on URL dataset obtained from the phishing site. This dataset consisted of 34,600 legitimate URLs’s and 38,149 falsified and fake URL’s. The dataset was further classified using the NLP approach.

Authors in [27] also contributed their work towards URL analysis and detection. The study focused on performing a comparative analysis by making use of four classifiers. However, the scholars made use of the dataset obtained from UCI repository. The study in [28] also implemented ML techniques to detect and classify legitimate files from that of Phishing attack files. The authors executed their work using ANN, SVM and Random Forest classifiers. It was observed that RF classifiers performed better and gave enhanced

results and accomplished an accuracy of 98.35 percent. UCI dataset repository was also used by authors in [29] to detect phishing attacks on websites using concepts of ML that were capable to extract and select 30 significant features, which were further trained and tested using six classifiers. The dataset comprised of 11,055 fake URL links.

B. Deep Learning based Detection Technique

Concepts of deep learning are also utilized in the area of intrusion detection systems. The basic idea of deep learning is to enact the same way as that of the human brain. Hence, its algorithm execution involves neurons. It was also witnessed by multiple researchers that the workflow of a neural network performed better than machine learning techniques and helped in accomplishing higher levels of accuracy. However, the implementation of neural network depends on various factors and learning parameters, making it a tedious task to execute. Deep learning algorithms generally include the working concept of CNN and RNN's. Further, the execution of detection and classification techniques using DL has proved to outperform the functioning of ML. Theoretically, this process occurs through the passage of input and vector outputs through layers. There are multiple layers in a conventional CNN along with hidden layers in the middle, which are responsible for the main classification and detection process. Below illustrated is the conventional workflow of a DL model for malware detection and classification.

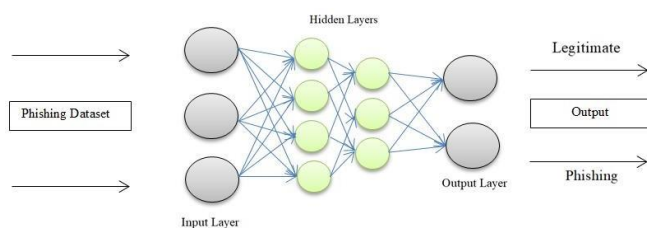


Fig. 3: Workflow of DL Classification

Authors in [30] worked upon pages on a website. They focused on designing a page layout for comparability factors to detect phishing attacks. To achieve this purpose, page phishing classifiers such as SVM and decision trees were used. In another study [31], authors contributed their research on a similar working pattern of conducting phishing page classifiers. However, in this case they conducted their experiments on more than 2 datasets that contained the links of 1530 phishing websites and details on payment gateway. A similar work of sending URL as an input was witnessed in [32] where features from HTML and URL were extracted and classified using a stacking model using NN and RF as classifiers.

Finally, in another survey [33], a comprehensive and detailed study on attack strategies and detection techniques were presented by authors. They also made use of feature extraction methods to accomplish higher accuracy.

V. CONCLUSION

The survey conducted in this research focuses on the existing methods of detecting phishing attacks and also puts forward the challenges faced by software developers in creating those techniques. The authors also witnessed the growing trends of such attacks by malware actors. Hence, the primary aim of this research work is to bring in a healthier

understanding of the working procedure of such attacks and create awareness amongst internet users. The review tends to create a holistic understanding of the overall framework of phishing attacks. The authors also compared various existing techniques by multiple scholars. It was further brought to notice that spear phishing and mail phishing attacks were the most common and widely used attacks by hackers to steal credential information of internet users. Therefore, the authors felt that there was a necessity to suggest detection techniques to avoid the overall situation of such attacks. The authors finally came up with ML and DL detection techniques that could further assist in reducing the number of attacks taking place on a legit site.

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